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Sugar detection using drop evaporation

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Abstract

Evaporation deposition of a spilt sugary drop on the supporting surface can attract ants to surround it. People have a long history of using this phenomenon as an implication of sugar in the drop. Unfortunately, it is hard to detect sugar concentration and has to depend exclusively on ants. Here, we show a facile strategy for the eye-naked detection on sugar concentrations in common liquid mixtures, based on their evaporation depositions. Our experiments show that evaporation drops without any sugar form clear ring-like depositions, and the width of the ring area enlarges with the increase in sugar concentration. We demonstrate that the increase in sugar concentration can increase the liquid viscosity and decrease the capillary flow velocity, thus weakening the "coffee ring" effect. Our further experiments indicate that the temperature has insignificant effects on the correlation between sugar concentrations and ring-like depositions, but the substrate wettability impacts on the correlation by promoting the formation of ring-like depositions. Based on the mechanism study, we develop a strategy for detecting sugar concentrations via quantitatively correlating them with the width of the ring area, and demonstrate that it is valid for various liquid mixtures, for example, carbonate beverage, liquid medicine, and plant nutrient. Our findings not only present new insights into the understanding of the sugary drop evaporation, but also provide a facile strategy of detecting sugar concentration that promises great applications in food safety, pharmaceutical detection, and agricultural product measurements.

INTRODUCTION

Rapid detection of sugar concentration is desired in many fields, for example, food safety, disease diagnosis, and agricultural production. Various detection strategies have been proposed, including spectroscopy,[1,2] enzyme-based glucose sensors,[3,4] electrochemical probes,[5,6] and colorimetric assays.[7–10] However, most strategies require very complex processes and/or specific equipment to separate sugar from the analyte systems, which makes detection user-unfriendly and have high risk of the secondary contamination. It is, therefore, necessary to propose a facile and effective method to detect the sugar concentration in liquid mixtures.

The "coffee ring" effect, which is manifested by a ring-like material deposition after drop evaporation, has attracted increasing attention. From the viewpoint of thermodynamics, the "coffee ring" effect is a heterogeneous mass transfer process driven by the non-uniform distribution of evaporation rate over a sessile drop. In the drop, the higher

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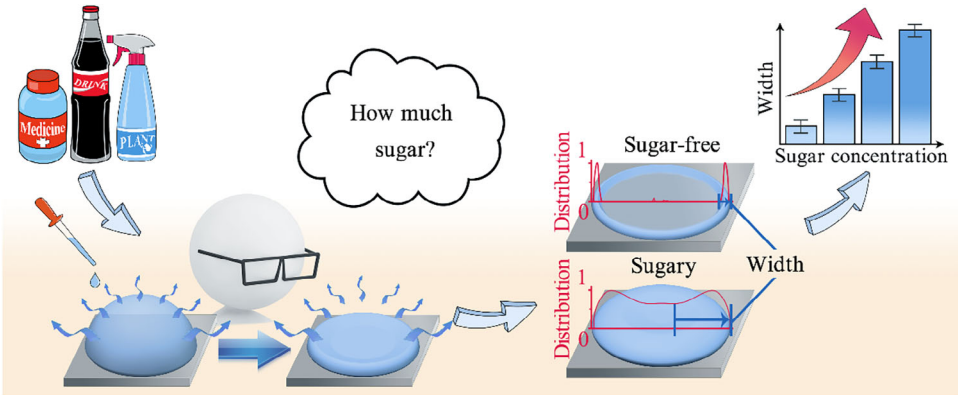

FIGURE 1 Sugar detection using drop evaporation. The schematics of the facile, eye-naked detection of the sugar concentration in liquid mixtures, simply by using their evaporation depositions; the detection is performed on the basis of the quantitative correlation between the width of the ring-like deposition and the sugar concentration, which demonstrates a wide range of application fields, for example, food safety, pharmaceutical engineering, and agricultural industry.

evaporation rate at the drop edge induces the outward capillary flow, which carries suspended materials from the center to edge, leading to the material deposition at the drop perimeter. Recently, the "coffee ring" effect has been reported to have a wide range of applications, for example, material self-separation,[11–13] sensor fabrication,[14,15] health diagnostics,[16,17] and nanofabrication of substrates for surface enhanced Raman spectroscopy.[18,19] All of them are taking advantages of the essential physics of the "coffee ring" effect, that is, material self-separation and non-uniform accumulation.[20–23] Given that, the "coffee ring" effect provides a potential methodology for realizing the rapid sugar detection.

Here, we propose a facile strategy to rapidly detect the sugar concentration, simply based on their evaporation depositions, as shown in Figure 1. Our experiments indicate that the increase in sugar concentration can enhance the liquid viscosity and decrease the capillary flow velocity, and thus weakening the "coffee ring" effect. We further find that the correlation between sugar concentrations and ring-like depositions can be influenced by the substrate wettability, albeit unaffected by the temperature. We develop a rapid sugar detection strategy via quantitatively correlating the width of ring-like depositions to sugar concentrations. This work provides a simple method for eye-naked detection on sugar concentrations for a wide range of applications, for example, food safety, pharmaceutical engineering, and agricultural industry.

RESULTS AND DISCUSSION

We first recorded the morphological evolution of cola drops with different sugar concentrations (C_s) versus the normalized evaporation time, which was calculated via dividing the evaporation time (t) by the total time elapsed for complete evaporation (t_f). The macroscopic observations are shown in Figure 2a. In the initial stage ($t/t_f < 2/3$), all cola drops exhibit the uniform distribution of non-volatile materials. When it comes to the second stage ($t/t_f \geq 2/3$), the non-volatile materials in sugar-free or low-sugar cola drop (40 and 80 mg/mL) start to aggregate at the drop perimeter to form the ring-like deposition, which subsequently develops from the edge to center with the propagation of drop evaporation; whereas, such a trend tends to be insignificant for the cola sessile drop with a high C_s (120 mg/mL). We calculated the normalized width of the ring-like deposition via dividing the real width (W) by the drop radius (r), that is, $W_r = W/r$, and quantitatively analyzed its time-dependent evolution process, as shown in Figure 2b. Interestingly, W_r in all cola drops begins to increase at $t/t_f \geq 2/3$, and the increase rate enhances significantly with C_s. We then determined the initial apparent contact angle of cola drops (θ_0) and recorded its time-dependent evolution process (θ_t), as shown in Figure 2c,d, respectively. Unexpectedly, there are insignificant differences in both of θ_0 and its evolution in all cola drops during evaporation. The apparent contact angle is intrinsically a macroscopic property of the wettability of a drop on a surface, and thus it only represents the macroscopic profile near the drop edge. However, the microscopic details of the drop in the second stage of evaporation, particularly material deposition at the contact line and material deposition in the central region, would be significantly different in cola drops with different C_s, which will be discussed in the next section.

We then observed the final evaporation depositions of cola drops and reconstructed their three-dimensional (3D) profiles by using an optical profilometer, as shown in Figure 3a. Clearly, in the sugar-free cola drop ($C_s = 0$ mg/mL), a very narrow ring-like deposition forms along the drop perimeter, which was driven by the "coffee ring" effect during evaporation. However, in the sugary cola drops, the width of the ring-like depositions increases with the increase in C_s, and reaches the threshold, that is, equal to the drop radius, when C_s is up to 120 mg/mL. Such an evolution trend was supported by observations by using a laser scanning confocal microscope, as shown in Figure 3a. In the meanwhile, the height of the profile view of the ring-like deposition increases with the increase in C_s, as shown in Figure 3b. The difference between the apex of the deposition in the edge region and the height of the central region decreases as C_s increases. All these experimental results

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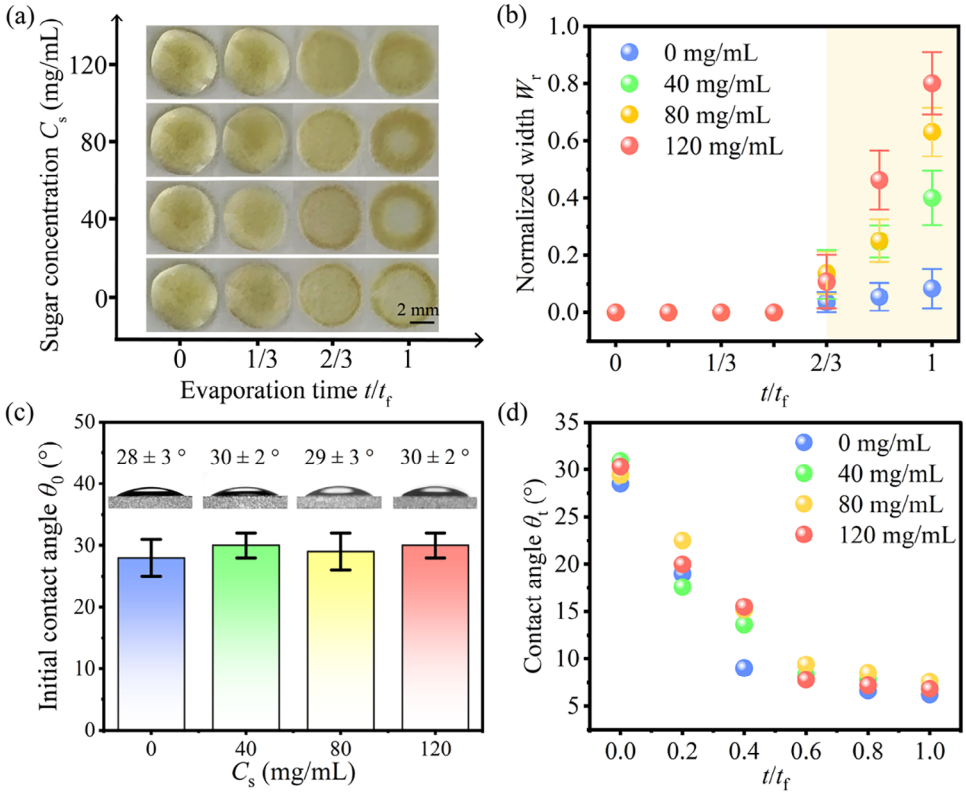

FIGURE 2 Evaporation of cola drops with different sugar concentrations. (a) Time-dependent morphological evolution of cola drops with sugar concentrations (C_s) at 0, 40, 80, and 120 mg/mL, respectively; the evaporation time was normalized via dividing the evaporation time (t) by the total time elapsed for complete evaporation (t/t_f). (b) Evolution of the normalized width of the ring-like deposition (W_r) during evaporation, based on the calculation of the morphological changes in (a). (c) Initial apparent contact angle (θ_0) of cola drops with different C_s. (d) Evolution of the apparent contact angle (θ_t) of cola drops during evaporation.

indicate that the existing sugar would weaken the "coffee ring" effect of the cola drop.

Considering the significant difference in the side view profile of the evaporation deposition, we further analyzed the arithmetic mean surface roughness (*Ra*) in the edge region and central region by using the atomic force microscope (AFM), as shown in Figure 3c. Obviously, the evaporation deposition of the sugar-free cola drop has a *Ra* that is much larger than those with sugar; furthermore, the difference in Ra between the edge region and central region significantly decreases with the increase in C_s. The microscopic observations on different regions of evaporation depositions (Figure 3d) indicate that there are many bubble-like deposits left by the evaporated sugar-free drop. These deposits could have a flattened surface, since they seem to be in a micrometer scale from two-dimensional scanning electron microscope (SEM) images (Figure 3d), but have *Ra* only in a nanometer scale (Figure 3c). In contrast, these deposits disappear in sugary cola drops (Figure 3d). Such a difference in these bubble-like deposits is the main contributor to the difference in *Ra* and could be related to the existence of sugar in the cola drop. The underlying mechanisms are, unfortunately, not very clear.

We next discuss the possible mechanisms of different evaporation depositions in cola drops with different C_s. From the point of view of the evaporation dynamics of the "coffee ring" effect, in an evaporating drop with a pinned contact line, the velocity of the hydrodynamic flow ($\bar{v}_r$) merely caused by the non-uniform distribution of evaporation flux (J_r) can be simply written as[20,21]:

$$\bar{v}_r \propto J_r \propto (1 - R_r)^{-\lambda(\theta)} \tag{1}$$

where R_r is the ratio of the radial distance to the radius of the drop and $\lambda(\theta)$ is a fitting parameter that is related to the apparent contact angle. However, in cola drops, C_s plays an insignificant role in influencing both initial apparent contact angle (θ_0) and its time-dependent evolution (θ_t), as shown in Figure 2c,d. Accordingly, $\bar{v}_r$ merely induced by J_r could, theoretically, be insignificantly different in these drops. We, therefore, suppose that there could be some intrinsic factors influencing the total flux of capillary flow (v), say from the wider perspective of the drop as a whole.

Given the above-mentioned assumption, we measured surface tension (γ) and density (ρ) of the sugar-free and sugary cola solutions with different C_s, as illustrated in Figure 4a,b, respectively. The increase in C_s from 0 to 120 mg/mL can increase γ from 63.18 ± 5.14 N/m to 80.10 ± 6.65 N/m and slightly increase ρ from 0.96 ± 0.01 g/mL to 0.99 ± 0.01 g/mL. We then determined the viscosity (η) of the sugar-free and sugary cola, as shown in Figure 4c, and found that the increase in C_s would significantly increase η from 0.0050 ± 0.0013 Pa s to

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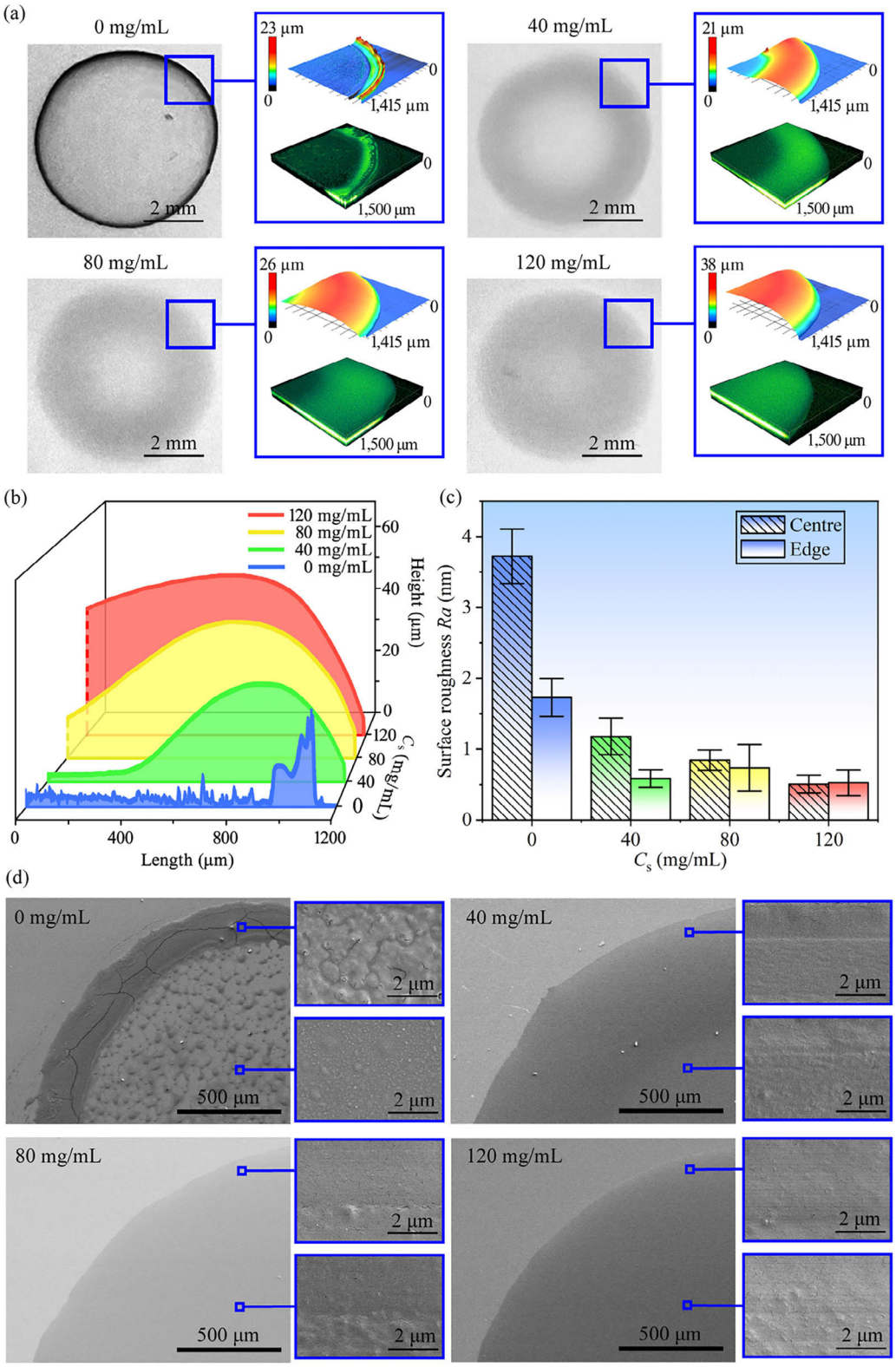

FIGURE 3 Evaporation depositions of cola drops. (a) Evaporation depositions of cola drops with different sugar concentrations, that is, $C_s = 0$, 40, 80, and 120 mg/mL; three-dimensional (3D) profiles were reconstructed by using the optical profilometer (colorful images) and the laser scanning confocal microscope (green images), respectively. (b) Side view profiles of evaporation depositions in (a). (c) Surface roughness of the evaporation deposition in the central region and edge region of the drop, which was analyzed by using the atomic force microscope (AFM). (d) Microscopic observations on the central region and edge region of evaporation depositions, which were using the scanning electron microscope (SEM), respectively.

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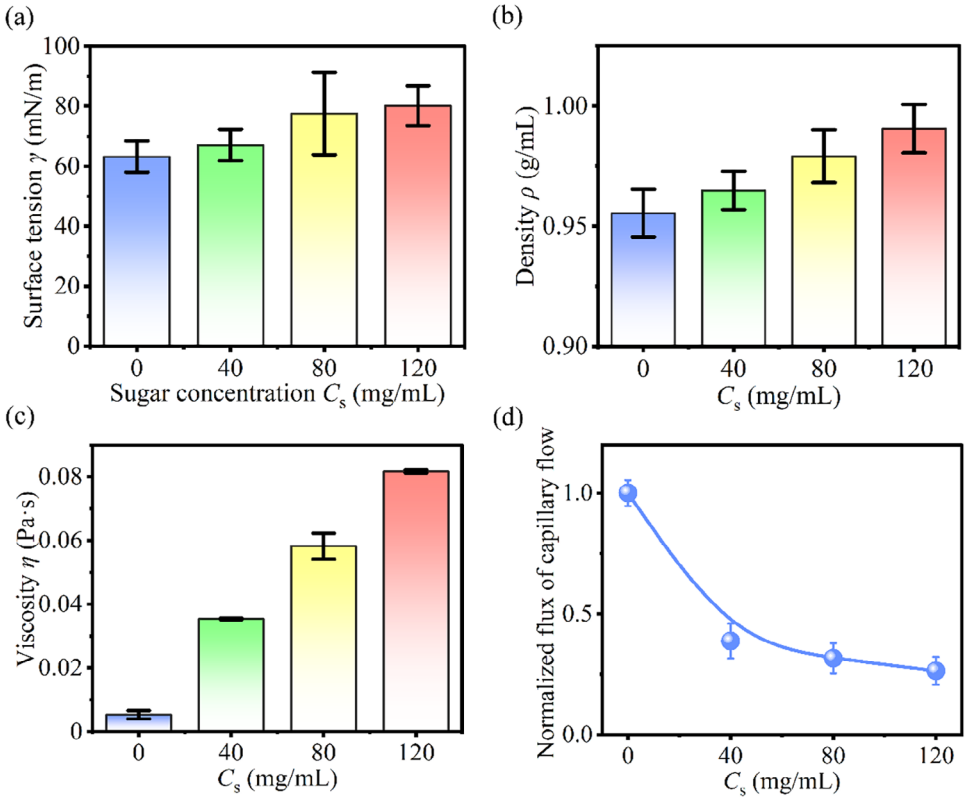

FIGURE 4 Mechanism understanding of effects of sugar concentrations on evaporation depositions. (a) Surface tension (γ) of sugar-free cola solutions and those with different sugar concentrations, that is, $C_s = 40$, 80, and 120 mg/mL, respectively. (b) Density (ρ) of sugar-free cola solutions and those with different C_s. (c) Effects of C_s on the viscosity (η) of cola solutions. (d) Relationship between the total flux of capillary flow as a function of diffusion coefficient ($\nu_{(D_\nu)}$) and C_s, which is calculated by substituting the experimental results of (a–c) into Equation (3).

0.0810 ± 0.0004 Pa s. ν can be written as[24]:

$$\nu \propto \sqrt{\frac{D_\nu \cdot P_\nu \cdot \gamma}{\eta \cdot \rho}} \tag{2}$$

where D_ν is the diffusion coefficient of suspended materials, which is inversely proportional to the liquid viscosity (η) according to Stokes–Einstein equation; P_ν is the saturated vapor pressure over the cola drop. With the further consideration of the Raoult's law, Equation (2) can be modified as:

$$\nu \propto \sqrt{\frac{D_\nu \cdot (1 - C_s) \cdot P_\nu^* \cdot \gamma}{\eta \cdot \rho}} \tag{3}$$

where P_ν^* is the saturated vapor pressure of the sugar-free cola drop, which can be assumed as a constant in this work. By coupling experimental results (Figure 4a–c) with the Equation (3), we established a simplified relationship between the total flux of capillary flow as a function of diffusion coefficient ($\nu_{(D_\nu)}$) and C_s; here, we set $\nu_{(D_\nu)}$ for sugar-free cola sessile drop as 1 and illustrate the results in Figure 4d. As C_s increases from 40 to 120 mg/mL, the normalized total flux of capillary flow decreases dramatically from 0.39 ± 0.07 to 0.26 ± 0.06; in fact, D_ν could also be decreased with the increase in C_s, and the coupling of these two aspects leads to the significant decrease in ν. Furthermore, as the evaporation propagates, C_s would further increase in sugary cola drops, along with the enlargement of η; as a result, ν could be continuously decreased, and the "coffee ring" effect could be further weakened by the concentrated sugar in the system. Our explanation only has the possible conceptual implication of the weakening of the "coffee ring" effect caused by the sugar in cola drop. In a practical situation, however, the drop evaporation includes more complex physical themes and underlying mechanisms should be much more complicated; the current understanding is still in a very preliminary stage.

On the other hand, we investigated the effects of temperature on evaporation depositions. As shown in Figure 5a,b, the temperature (10°C and 40°C) had insignificant effects on the correlation between ring-like depositions and sugar concentrations (0–120 mg/mL). Furthermore, we studied evaporation depositions of cola drops on different substrates with different θ_0, as shown in Figure 5c and Supporting Information: Figure S1. Clearly, the variation ranges of sugar concentrations related to the ring-like depositions reduce with the decrease in the wettability of the drop on the substrate, that is, 0–30 mg/mL when θ_0 is ~77°. We note that there is no "coffee ring" effect for the cola drop on a hydrophobic surface ($\theta_0 \sim 145°$), as shown in Supporting Information: Figure S1.

We now explore the potential of using evaporation depositions of sugar-free and sugary cola drops as a tool to detect the sugar

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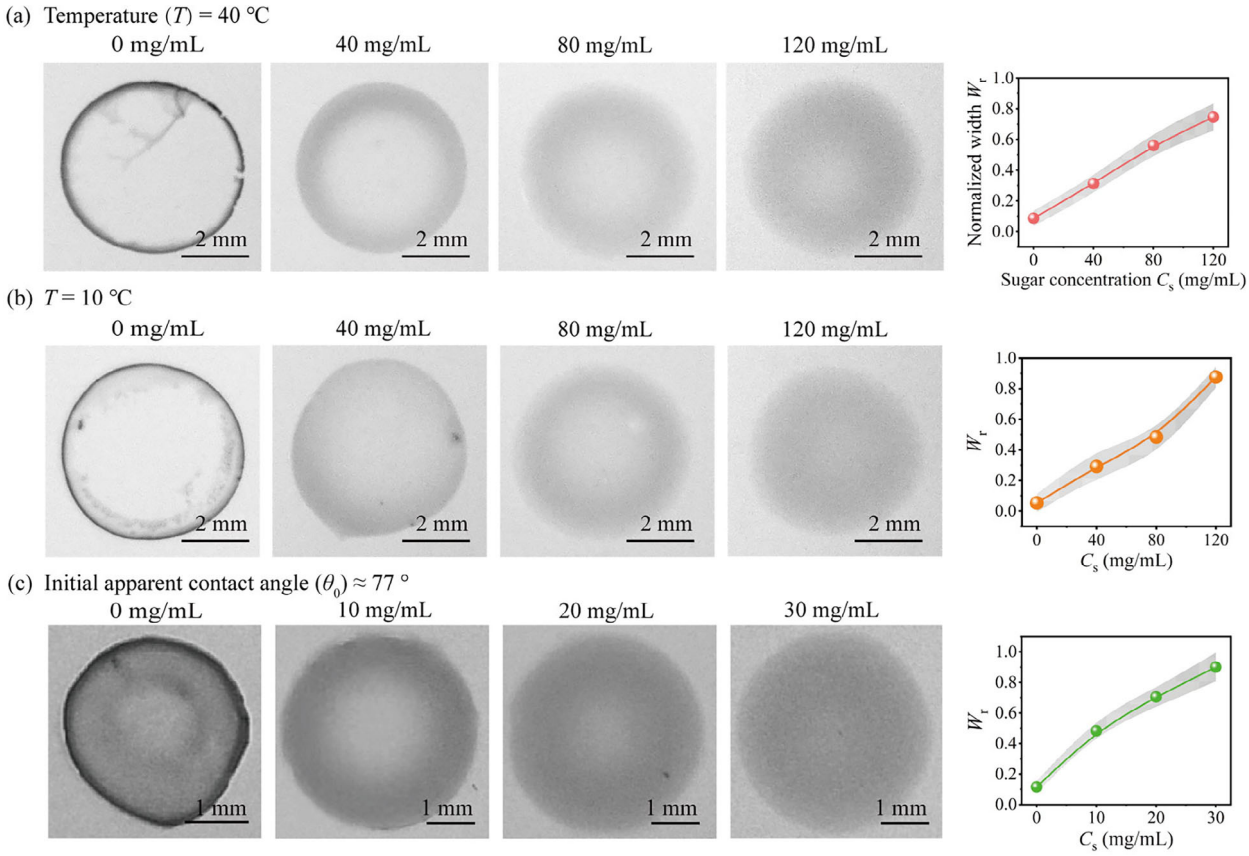

FIGURE 5 Factors influencing evaporation depositions related to sugar concentrations. Evaporation depositions of cola drops with different sugar concentrations, that is, $C_s = 0, 40, 80$, and 120 mg/mL, and the calculation of W_r of cola drops evaporating at temperature of (a) 40°C and (b) 10°C. (c) Evaporation depositions of cola drops on plastic substrate with $\theta_0 \sim 77°$.

concentration. Based on the results of the factors analysis, we set the evaporation temperature to 25°C and used glass slides as the substrate. Subsequently, we established the quantitative relationship between W_r and C_s, based on at least six individual groups of evaporated cola drops. As illustrated in Figure 6a, W_r is positively associated with C_s in a fitting equation as:

$$W_r = 0.07 + 10^{-2}C_s - 3.82 \times 10^{-5}C_s^2 \quad (4)$$

The coefficient of determination (R^2) can be written as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - f_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (5)$$

in which n is the number of data set, y_i is the observed data of W_r, f_i is the fitted value of W_r in Equation (4), $\bar{y}$ is the means of observed data of W_r. The value of R^2 is 0.98, showing the high reliability of the evaporation depositions on sugar indication. To demonstrate the universality of this detection strategy, we selected another two common liquid mixtures, that is, liquid stomach medicine (Figure 6b,c) and plant nutrient (Figure 6d,e), to analyze the relationship between C_s and W_r of their evaporation depositions. As expected, W_r of the evaporated drop of liquid stomach medicine shows the positive correlation with C_s (Figure 6c,e) in a fitting equation as:

$$W_r = 0.20 + 2.02 \times 10^{-5}C_s^2 \quad (6)$$

in which $R^2 = 0.95$. We note that the similar trend was also found in the drop of plant nutrient with different C_s, fitting well with the equation as:

$$W_r = 0.09 + 1.46 \times 10^{-5}C_s^2 \quad (7)$$

where $R^2 = 0.98$. All these experimental results suggest that the positive correlation between the width of the ring-like deposition of an evaporated drop and its sugar concentration exists widely in various liquid mixtures, and thus holding promise to be as a rapid sugar detection tool with a wide range of application fields.

CONCLUSION

In summary, we proposed a facile, effective strategy for rapidly detecting sugar concentrations in common liquid systems, for example, carbonated beverages, liquid medicine, and plant nutrient, by simply

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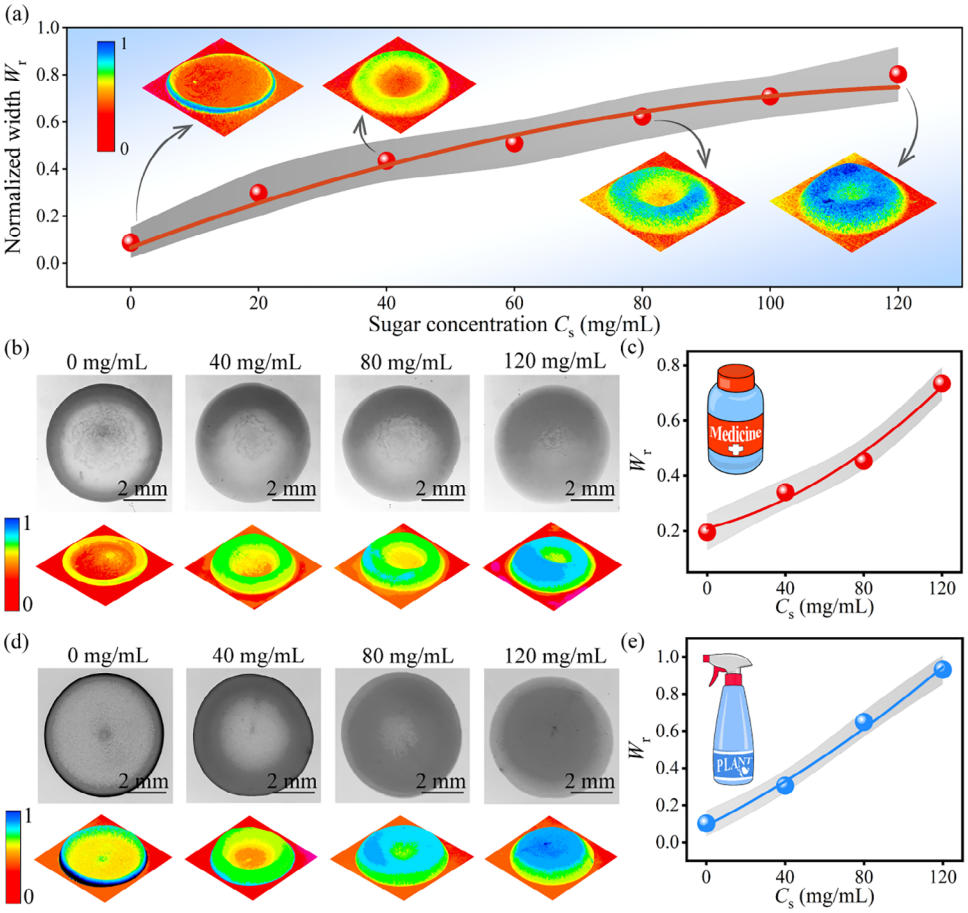

FIGURE 6 Detection of sugar concentrations in common liquid mixtures using drop evaporation. (a) Calculation of the normalized width of the ring-like deposition (W_r) of cola drops with different C_s (0, 40, 80, and 120 mg/mL), clearly showing that W_r increases with the increase in C_s; insets are colorful 3D images of evaporation depositions. (b) Evaporation depositions of liquid stomach medicine drops with different C_s. (c) Calculation of W_r of liquid stomach medicine drops. (d) Evaporation depositions of plant nutrient drops with different C_s. (e) Calculation of W_r of plant nutrient drops.

using their evaporation depositions. This strategy was developed on the basis of our experimental observations that the width of the ring-like deposition left by the drop evaporation had a positive correlation with initial sugar concentrations. Our further experiments and analysis indicated that the increase in sugar concentrations decreased the velocity of the capillary flow within the drop via influencing its intrinsic properties, including viscosity, density, and surface tension, which would weaken the "coffee ring" effect, and thus enlarging the ring-like depositions after drop evaporation. We further found that the temperature had insignificant effects on the correlation between sugar concentration and ring-like depositions, but the wettability of the drop on the substrate promoted the formation of ring-like depositions. We also developed a highly reliable strategy of sugar detection by establishing the quantitative correlation between the width of the ring-like deposition and sugar concentrations in common liquid mixtures. We note that this strategy should have a wide range of applications, which are not restricted in fields of food safety, pharmaceutical detection, and agricultural product measurements. On the other hand, the theoretical basis as proposed would help to unlock the mystery of some natural ubiquitous phenomena, for example, circular aggregation of ants surrounding a spilt evaporated sugary drop, from the point of view of the "coffee ring" effect. Our findings not only present new insights into the understanding of the evaporation dynamics in a sugary drop, but also provide a simple methodology of sugar concentration detection for various application fields.

METHODS

Sugar-free cola from Coca-Cola and the white sugar from Socona were used in this work. The CONBA sugar-free stomach medicine was purchased from Ali Pharmacy and the Flower Master sugar-free plant nutrient was obtained from flower shop. The ingredients of sugar-free cola, sugar-free stomach medicine, and sugar-free plant nutrient are shown in Supporting Information PS2. All chemicals were of analytical grade and used without any further purification. Millipore water (18.2 MΩ cm) was prepared by using a laboratory pure water system (ak Exceed-Cb-10). Glass slides from Objektträger (Germany) were rinsed using ethanol (Ali Pharmacy) and ultrapure water consecutively. Subsequently, these glass slides were thermally dried in an oven at a

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constant temperature of 40°C (JINGHONG Y802) for 30 min. They were used within 1 h.

Evaporation of cola drops was conducted by the following steps. Sugar solutions with varying concentrations (40, 80, and 120 mg/mL) were prepared by dissolving white sugar in sugar-free cola and mixing on a magnetic stirrer (LICHEN LC-MSA-HD) at room temperature for 1 h. A 15 ± 0.1 μL drop of each solution was carefully deposited onto an ultraclean glass slide using an Eppendorf pipette and allowed to evaporate under controlled conditions (25.0 ± 0.5°C, 60 ± 10% relative humidity).

The ultraclean glass slides were preheated in biochemical incubators (0°C–60°C, SPT-P150C) at 40°C and 10°C, respectively, for 30 min. Subsequently, 15 μL of sugar-free or sugary cola drop was carefully placed onto the glass slides with different temperatures using an Eppendorf pipette and evaporated naturally at 40.0 ± 1.0°C and 10.0 ± 1.0°C, respectively. The relative humidity was 60 ± 10%. The super-hydrophobic glass slide was made by immersing the ultraclean glass in the nano-silicon dioxide solution for 3 s, then placed at room temperature until the solvent completely volatilized. The preparation method of the nano-silicon dioxide solution is provided in Supporting Information PS3. Then, 15 μL of sugar-free or sugary cola drop was gently placed onto the different substrate, that is, plastic petri dish (BKMAN), super-hydrophobic glass, by using an Eppendorf pipette, and evaporated naturally at controlled conditions (25.0 ± 0.5°C, 60 ± 10% relative humidity).

1.5 g of stomach medicine was ground into fine powders in a mortar (LICHEN). Liquid stomach medicine was prepared by mixing 1.5 g of stomach medicine powder with 50 mL of water. The liquid stomach medicine with different sugar concentrations (i.e., 40, 80, and 120 mg/mL) was prepared by dissolving the white sugar into the liquid stomach medicine. The plant nutrient with different sugar concentrations (i.e., 40, 80, and 120 mg/mL) was prepared by dissolving the white sugar into the plant nutrient. One drop of sugar-free or sugary solutions (15 μL) was gently placed onto the ultraclean glass surface by using an Eppendorf pipette, and let it evaporate naturally at a controlled condition, which remains consistent with that of cola drops.

To characterize the morphology of evaporated depositions, the morphological evolution was recorded by using a Samsung mobile phone, and the morphological details were observed by using an optical microscope (CNOPTEC MIT500). Three-dimensional profile of the evaporated deposition was reconstructed by using a Contour GT-K 3D optical profilometer (Bruker). The distribution of suspended materials after drop evaporation was also observed by a Nikon A1 confocal laser scanning microscope. The surface roughness of both central and edge regions of the evaporated deposition was analyzed by using a Dimension Icon 5000 (Bruker) AFM, with non-contact mode. The central region refers to the area near the drop center, and the edge region refers to the ring-shaped area. The microscopic observations were performed by using SU8010 (Hitachi) SEM. The viscosity of the cola solutions with different sugar concentrations was measured by using a Ubbelohde viscometer. The surface tension was determined by using the pendant drop method. The quantitative analysis on physical properties of the evaporated deposition, that is, the width of the ring-like deposition and the apparent contact angle of the drop, were performed by using the Origin software, coupled with the ImageJ software. The methods for measuring the width of the ring-like deposition and constructing 3D images by using the ImageJ were shown in Supporting Information PS4 and PS5. All the statistic information was obtained based on at least six individual experimental groups to ensure the reliability and repeatability of the experimental results.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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